

1000 AI Projects — Foundation to GenAI & Agentic AI

LLM · UI (React / Streamlit) · Backend (Python FastAPI) · AI Frameworks (LangGraph / LangChain / AutoGen / CrewAI / LlamaIndex / DSPy) · Vector DB & Infra

Master in AI Engineering — Build production-grade GenAI, Agentic AI, RAG & MLOps systems.
From LLM foundations to multi-agent orchestration — the complete hands-on path for AI engineers.

Created By Balawant M Kadam
Exported: 20/9/2026, 9:05:53 pm
Projects in export: 180 of 1000
Filters: Level: intermediate | Type: All | Domain: All | Framework: All

#0271

TRADITIONAL

INTERMEDIATE

Social Listening RAG Assistant

Marketing & Advertising

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over social listening, exposed through a typed API with tracing, caching and evaluation.

LLM	Gemini 1.5 Pro
UI	React + Vite + Tailwind
BACKEND	Python · FastAPI + MinIO / S3
AI FRAMEWORKS	LangChain FAISS FastAPI
VECTOR / INFRA	Pinecone · DuckDB

#0272

TRADITIONAL

INTERMEDIATE

Attribution Modeling Agent

Marketing & Advertising

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over attribution modeling, exposed through a typed API with tracing, caching and evaluation.

LLM	Llama 3.3 70B
UI	Streamlit + FastAPI
BACKEND	Python · FastAPI + APScheduler
AI FRAMEWORKS	LangGraph LangChain ChromaDB Docker OpenAI Agents SDK
VECTOR / INFRA	Weaviate · LanceDB

#0273

TRADITIONAL

INTERMEDIATE

Ticket Classification Copilot

Customer Support

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over ticket classification, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LlamaIndex Qdrant LangSmith
VECTOR / INFRA	Qdrant · OpenSearch

#0274

TRADITIONAL

INTERMEDIATE

Virtual Support Agent Pipeline

Customer Support

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over virtual support agent, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	DSPy LlamaIndex pgvector LangFuse FAISS
VECTOR / INFRA	Milvus · FAISS

#0275

TRADITIONAL

INTERMEDIATE

Sentiment Routing Knowledge Base

Customer Support

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over sentiment routing, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LangChain MLflow Pydantic
VECTOR / INFRA	pgvector · ChromaDB

#0276

TRADITIONALINTERMEDIATE

Knowledge Base Search Workflow Automation

Customer Support

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over knowledge base search, exposed through a typed API with tracing, caching and evaluation.

LLM	Claude 3.5 Sonnet
UI	React + TanStack Query
BACKEND	Python · FastAPI + SQLAlchemy 2.0 + Alembic
AI FRAMEWORKS	LangGraphLangChainFastAPIPydantic AIQdrant
VECTOR / INFRA	Neo4j · Pinecone

#0277

TRADITIONALINTERMEDIATE

Escalation Prediction Evaluation Harness

Customer Support

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over escalation prediction, exposed through a typed API with tracing, caching and evaluation.

LLM	Gemini 1.5 Pro
UI	React + Vite + Tailwind
BACKEND	Python · FastAPI + MinIO / S3
AI FRAMEWORKS	LlamaIndexDockerOpenAI Agents SDK
VECTOR / INFRA	Elasticsearch · Weaviate

#0278

TRADITIONALINTERMEDIATE

Voice Support Bot API Service

Customer Support

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over voice support bot, exposed through a typed API with tracing, caching and evaluation.

LLM	Llama 3.3 70B
UI	Streamlit + FastAPI
BACKEND	Python · FastAPI + APScheduler
AI FRAMEWORKS	DSPyLlamaIndexLangSmithPEFT / LoRAMLflow
VECTOR / INFRA	DuckDB · Qdrant

#0279

TRADITIONAL

INTERMEDIATE

Contract Review RAG Assistant

Legal & Compliance

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over contract review, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LangChain LangFuse FAISS
VECTOR / INFRA	LanceDB · Milvus

#0280

TRADITIONAL

INTERMEDIATE

Legal Research Agent

Legal & Compliance

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over legal research, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	LangGraph LangChain Pydantic ChromaDB Docker
VECTOR / INFRA	OpenSearch · pgvector

#0281

TRADITIONAL

INTERMEDIATE

Policy Compliance Copilot

Legal & Compliance

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over policy compliance, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LlamaIndex Pydantic AI Qdrant
VECTOR / INFRA	FAISS · Neo4j

#0282

TRADITIONAL

INTERMEDIATE

E-Discovery Pipeline

Legal & Compliance

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over e-discovery, exposed through a typed API with tracing, caching and evaluation.

LLM	Claude 3.5 Sonnet
UI	React + TanStack Query
BACKEND	Python · FastAPI + SQLAlchemy 2.0 + Alembic
AI FRAMEWORKS	DSPy LlamaIndex OpenAI Agents SDK pgvector LangFuse
VECTOR / INFRA	ChromaDB · Elasticsearch

#0283

TRADITIONAL

INTERMEDIATE

Clause Extraction Knowledge Base

Legal & Compliance

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over clause extraction, exposed through a typed API with tracing, caching and evaluation.

LLM	Gemini 1.5 Pro
UI	React + Vite + Tailwind
BACKEND	Python · FastAPI + MinIO / S3
AI FRAMEWORKS	LangChain PEFT / LoRA MLflow
VECTOR / INFRA	Pinecone · DuckDB

#0284

TRADITIONAL

INTERMEDIATE

Regulatory Monitoring Workflow Automation

Legal & Compliance

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over regulatory monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM	Llama 3.3 70B
UI	Streamlit + FastAPI
BACKEND	Python · FastAPI + APScheduler
AI FRAMEWORKS	LangGraph LangChain FAISS FastAPI Pydantic AI
VECTOR / INFRA	Weaviate · LanceDB

#0285

TRADITIONAL

INTERMEDIATE

Resume Screening Evaluation Harness

Human Resources

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over resume screening, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LlamaIndex ChromaDB Docker
VECTOR / INFRA	Qdrant · OpenSearch

#0286

TRADITIONAL

INTERMEDIATE

Interview Assistant API Service

Human Resources

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over interview assistant, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	DSPy LlamaIndex Qdrant LangSmith PEFT / LoRA
VECTOR / INFRA	Milvus · FAISS

#0287

TRADITIONAL

INTERMEDIATE

Employee Onboarding RAG Assistant

Human Resources

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over employee onboarding, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LangChain pgvector LangFuse
VECTOR / INFRA	pgvector · ChromaDB

#0288

TRADITIONAL

INTERMEDIATE

Attrition Prediction Agent

Human Resources

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over attrition prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS

LangGraph

LangChain

MLflow

Pydantic

ChromaDB

VECTOR / INFRA

Neo4j · Pinecone

#0289

TRADITIONAL

INTERMEDIATE

Skills Gap Analysis Copilot

Human Resources

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over skills gap analysis, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS

LlamaIndex

FastAPI

Pydantic AI

VECTOR / INFRA

Elasticsearch · Weaviate

#0290

TRADITIONAL

INTERMEDIATE

Performance Review Summarizer Pipeline

Human Resources

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over performance review summarizer, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS

DSPy

LlamaIndex

Docker

OpenAI Agents SDK

pgvector

VECTOR / INFRA

DuckDB · Qdrant

#0291

TRADITIONAL

INTERMEDIATE

Crop Disease Detection Knowledge Base

Agriculture

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over crop disease detection, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LangChain LangSmith PEFT / LoRA
VECTOR / INFRA	LanceDB · Milvus

#0292

TRADITIONAL

INTERMEDIATE

Yield Prediction Workflow Automation

Agriculture

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over yield prediction, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	LangGraph LangChain LangFuse FAISS FastAPI
VECTOR / INFRA	OpenSearch · pgvector

#0293

TRADITIONAL

INTERMEDIATE

Precision Irrigation Evaluation Harness

Agriculture

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over precision irrigation, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LlamaIndex Pydantic ChromaDB
VECTOR / INFRA	FAISS · Neo4j

#0294

TRADITIONAL

INTERMEDIATE

Pest Monitoring API Service

Agriculture

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over pest monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Pydantic AI](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0295

TRADITIONAL

INTERMEDIATE

Soil Health Analytics RAG Assistant

Agriculture

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over soil health analytics, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA Pinecone · DuckDB

#0296

TRADITIONAL

INTERMEDIATE

Farm Drone Imagery Agent

Agriculture

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over farm drone imagery, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [PEFT / LoRA](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA Weaviate · LanceDB

#0297

TRADITIONAL

INTERMEDIATE

Grid Load Forecasting Copilot

Energy & Utilities

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over grid load forecasting, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA Qdrant · OpenSearch

#0298

TRADITIONAL

INTERMEDIATE

Renewable Output Prediction Pipeline

Energy & Utilities

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over renewable output prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [ChromaDB](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA Milvus · FAISS

#0299

TRADITIONAL

INTERMEDIATE

Asset Health Monitoring Knowledge Base

Energy & Utilities

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over asset health monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA pgvector · ChromaDB

#0300

TRADITIONAL

INTERMEDIATE

Energy Trading Workflow Automation

Energy & Utilities

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over energy trading, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [pgvector](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA Neo4j · Pinecone

#0301

TRADITIONAL

INTERMEDIATE

Outage Detection Evaluation Harness

Energy & Utilities

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over outage detection, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0302

TRADITIONAL

INTERMEDIATE

Smart Meter Analytics API Service

Energy & Utilities

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over smart meter analytics, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [FastAPI](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA DuckDB · Qdrant

#0303

TRADITIONAL

INTERMEDIATE

Citizen Service Assistant RAG Assistant

Government & Public Sector

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over citizen service assistant, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA LanceDB · Milvus

#0304

TRADITIONAL

INTERMEDIATE

Document Processing Agent

Government & Public Sector

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over document processing, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [LangSmith](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA OpenSearch · pgvector

#0305

TRADITIONAL

INTERMEDIATE

Policy Impact Analysis Copilot

Government & Public Sector

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over policy impact analysis, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA FAISS · Neo4j

#0306

TRADITIONAL

INTERMEDIATE

Emergency Response Pipeline

Government & Public Sector

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over emergency response, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Pydantic](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0307

TRADITIONAL

INTERMEDIATE

Permit Review Knowledge Base

Government & Public Sector

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over permit review, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA Pinecone · DuckDB

#0308

TRADITIONAL

INTERMEDIATE

Public Feedback Mining Workflow Automation

Government & Public Sector

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over public feedback mining, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [OpenAI Agents SDK](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA Weaviate · LanceDB

#0309

TRADITIONAL

INTERMEDIATE

Content Recommendation Evaluation Harness

Media & Entertainment

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over content recommendation, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS

LlamaIndex

PEFT / LoRA

MLflow

VECTOR / INFRA Qdrant · OpenSearch

#0310

TRADITIONAL

INTERMEDIATE

Script Generation API Service

Media & Entertainment

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over script generation, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS

DSPy

LlamaIndex

FAISS

FastAPI

Pydantic AI

VECTOR / INFRA Milvus · FAISS

#0311

TRADITIONAL

INTERMEDIATE

Video Editing Assistant RAG Assistant

Media & Entertainment

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over video editing assistant, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS

LangChain

ChromaDB

Docker

VECTOR / INFRA pgvector · ChromaDB

#0312

TRADITIONAL

INTERMEDIATE

Music Generation Agent

Media & Entertainment

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over music generation, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Qdrant](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA Neo4j · Pinecone

#0313

TRADITIONAL

INTERMEDIATE

Subtitle Localization Copilot

Media & Entertainment

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over subtitle localization, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0314

TRADITIONAL

INTERMEDIATE

Content Moderation Pipeline

Media & Entertainment

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over content moderation, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [MLflow](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA DuckDB · Qdrant

#0315

TRADITIONAL

INTERMEDIATE

NPC Dialogue Systems Knowledge Base

Gaming

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over npc dialogue systems, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA LanceDB · Milvus

#0316

TRADITIONAL

INTERMEDIATE

Procedural Level Generation Workflow Automation

Gaming

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over procedural level generation, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Docker](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA OpenSearch · pgvector

#0317

TRADITIONAL

INTERMEDIATE

Player Behavior Analytics Evaluation Harness

Gaming

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over player behavior analytics, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA FAISS · Neo4j

#0318

TRADITIONAL

INTERMEDIATE

Matchmaking API Service

Gaming

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over matchmaking, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [LangFuse](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0319

TRADITIONAL

INTERMEDIATE

Anti-Cheat Detection RAG Assistant

Gaming

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over anti-cheat detection, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Pinecone · DuckDB

#0320

TRADITIONAL

INTERMEDIATE

Game Testing Bots Agent

Gaming

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over game testing bots, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Pydantic AI](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA Weaviate · LanceDB

#0321

TRADITIONAL

INTERMEDIATE

Pick-and-Place Control Copilot

Robotics

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over pick-and-place control, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS LlamaIndex OpenAI Agents SDK pgvector

VECTOR / INFRA Qdrant · OpenSearch

#0322

TRADITIONAL

INTERMEDIATE

SLAM Navigation Pipeline

Robotics

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over slam navigation, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS DSPy LlamaIndex PEFT / LoRA MLflow Pydantic

VECTOR / INFRA Milvus · FAISS

#0323

TRADITIONAL

INTERMEDIATE

Manipulation Planning Knowledge Base

Robotics

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over manipulation planning, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS LangChain FAISS FastAPI

VECTOR / INFRA pgvector · ChromaDB

#0324

TRADITIONAL

INTERMEDIATE

Human-Robot Collaboration Workflow Automation

Robotics

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over human-robot collaboration, exposed through a typed API with tracing, caching and evaluation.

LLM	Claude 3.5 Sonnet
UI	React + TanStack Query
BACKEND	Python · FastAPI + SQLAlchemy 2.0 + Alembic
AI FRAMEWORKS	LangGraph LangChain ChromaDB Docker OpenAI Agents SDK
VECTOR / INFRA	Neo4j · Pinecone

#0325

TRADITIONAL

INTERMEDIATE

Swarm Coordination Evaluation Harness

Robotics

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over swarm coordination, exposed through a typed API with tracing, caching and evaluation.

LLM	Gemini 1.5 Pro
UI	React + Vite + Tailwind
BACKEND	Python · FastAPI + MinIO / S3
AI FRAMEWORKS	LlamaIndex Qdrant LangSmith
VECTOR / INFRA	Elasticsearch · Weaviate

#0326

TRADITIONAL

INTERMEDIATE

Sim-to-Real Transfer API Service

Robotics

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over sim-to-real transfer, exposed through a typed API with tracing, caching and evaluation.

LLM	Llama 3.3 70B
UI	Streamlit + FastAPI
BACKEND	Python · FastAPI + APScheduler
AI FRAMEWORKS	DSPy LlamaIndex pgvector LangFuse FAISS
VECTOR / INFRA	DuckDB · Qdrant

#0327

TRADITIONAL

INTERMEDIATE

Perception Fusion RAG Assistant

Autonomous Vehicles

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over perception fusion, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA LanceDB · Milvus

#0328

TRADITIONAL

INTERMEDIATE

Trajectory Prediction Agent

Autonomous Vehicles

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over trajectory prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [FastAPI](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA OpenSearch · pgvector

#0329

TRADITIONAL

INTERMEDIATE

Driver Monitoring Copilot

Autonomous Vehicles

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over driver monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA FAISS · Neo4j

#0330

TRADITIONAL

INTERMEDIATE

HD Map Generation Pipeline

Autonomous Vehicles

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over hd map generation, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [LangSmith](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0331

TRADITIONAL

INTERMEDIATE

Scenario Simulation Knowledge Base

Autonomous Vehicles

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over scenario simulation, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA Pinecone · DuckDB

#0332

TRADITIONAL

INTERMEDIATE

Fleet Teleoperation Workflow Automation

Autonomous Vehicles

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over fleet teleoperation, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Pydantic](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA Weaviate · LanceDB

#0333

TRADITIONAL

INTERMEDIATE

Literature Synthesis Evaluation Harness

Scientific Research

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over literature synthesis, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LlamaIndex Pydantic AI Qdrant
VECTOR / INFRA	Qdrant · OpenSearch

#0334

TRADITIONAL

INTERMEDIATE

Protein Structure Prediction API Service

Scientific Research

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over protein structure prediction, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	DSPy LlamaIndex OpenAI Agents SDK pgvector LangFuse
VECTOR / INFRA	Milvus · FAISS

#0335

TRADITIONAL

INTERMEDIATE

Experiment Design RAG Assistant

Scientific Research

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over experiment design, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LangChain PEFT / LoRA MLflow
VECTOR / INFRA	pgvector · ChromaDB

#0336

TRADITIONAL

INTERMEDIATE

Lab Automation Agent

Scientific Research

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over lab automation, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [FAISS](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA Neo4j · Pinecone

#0337

TRADITIONAL

INTERMEDIATE

Materials Discovery Copilot

Scientific Research

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over materials discovery, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0338

TRADITIONAL

INTERMEDIATE

Research Data Analysis Pipeline

Scientific Research

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over research data analysis, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Qdrant](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA DuckDB · Qdrant

#0339

TRADITIONAL

INTERMEDIATE

Emissions Monitoring Knowledge Base

Climate & Environment

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over emissions monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LangChain pgvector LangFuse
VECTOR / INFRA	LanceDB · Milvus

#0340

TRADITIONAL

INTERMEDIATE

Wildfire Prediction Workflow Automation

Climate & Environment

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over wildfire prediction, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	LangGraph LangChain MLflow Pydantic ChromaDB
VECTOR / INFRA	OpenSearch · pgvector

#0341

TRADITIONAL

INTERMEDIATE

Biodiversity Tracking Evaluation Harness

Climate & Environment

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over biodiversity tracking, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LlamaIndex FastAPI Pydantic AI
VECTOR / INFRA	FAISS · Neo4j

#0342

TRADITIONAL

INTERMEDIATE

Carbon Accounting API Service

Climate & Environment

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over carbon accounting, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Docker](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0343

TRADITIONAL

INTERMEDIATE

Weather Forecasting RAG Assistant

Climate & Environment

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over weather forecasting, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA Pinecone · DuckDB

#0344

TRADITIONAL

INTERMEDIATE

Ocean Data Analytics Agent

Climate & Environment

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over ocean data analytics, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [LangFuse](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA Weaviate · LanceDB

#0345

TRADITIONAL

INTERMEDIATE

Network Optimization Copilot

Telecommunications

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over network optimization, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Qdrant · OpenSearch

#0346

TRADITIONAL

INTERMEDIATE

Churn Prediction Pipeline

Telecommunications

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over churn prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Pydantic AI](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA Milvus · FAISS

#0347

TRADITIONAL

INTERMEDIATE

Fault Diagnosis Knowledge Base

Telecommunications

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over fault diagnosis, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA pgvector · ChromaDB

#0348

TRADITIONAL

INTERMEDIATE

Customer Care Assistant Workflow Automation

Telecommunications

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over customer care assistant, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [PEFT / LoRA](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA Neo4j · Pinecone

#0349

TRADITIONAL

INTERMEDIATE

5G Slice Management Evaluation Harness

Telecommunications

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over 5g slice management, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0350

TRADITIONAL

INTERMEDIATE

Spectrum Analytics API Service

Telecommunications

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over spectrum analytics, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [ChromaDB](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA DuckDB · Qdrant

#0351

TRADITIONAL

INTERMEDIATE

Property Valuation RAG Assistant

Real Estate & Construction

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over property valuation, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LangChain Qdrant LangSmith
VECTOR / INFRA	LanceDB · Milvus

#0352

TRADITIONAL

INTERMEDIATE

Site Safety Monitoring Agent

Real Estate & Construction

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over site safety monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	LangGraph LangChain pgvector LangFuse FAISS
VECTOR / INFRA	OpenSearch · pgvector

#0353

TRADITIONAL

INTERMEDIATE

Lease Abstraction Copilot

Real Estate & Construction

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over lease abstraction, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LlamaIndex MLflow Pydantic
VECTOR / INFRA	FAISS · Neo4j

#0354

TRADITIONAL

INTERMEDIATE

BIM Analysis Pipeline

Real Estate & Construction

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over bim analysis, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS

DSPy

LlamaIndex

FastAPI

Pydantic AI

Qdrant

VECTOR / INFRA ChromaDB · Elasticsearch

#0355

TRADITIONAL

INTERMEDIATE

Construction Scheduling Knowledge Base

Real Estate & Construction

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over construction scheduling, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS

LangChain

Docker

OpenAI Agents SDK

VECTOR / INFRA Pinecone · DuckDB

#0356

TRADITIONAL

INTERMEDIATE

Tenant Support Workflow Automation

Real Estate & Construction

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over tenant support, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS

LangGraph

LangChain

LangSmith

PEFT / LoRA

MLflow

VECTOR / INFRA Weaviate · LanceDB

#0357

TRADITIONAL

INTERMEDIATE

Itinerary Planning Evaluation Harness

Travel & Hospitality

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over itinerary planning, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS LlamaIndex LangFuse FAISS

VECTOR / INFRA Qdrant · OpenSearch

#0358

TRADITIONAL

INTERMEDIATE

Dynamic Pricing API Service

Travel & Hospitality

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over dynamic pricing, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS DSPy LlamaIndex Pydantic ChromaDB Docker

VECTOR / INFRA Milvus · FAISS

#0359

TRADITIONAL

INTERMEDIATE

Review Analytics RAG Assistant

Travel & Hospitality

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over review analytics, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS LangChain Pydantic AI Qdrant

VECTOR / INFRA pgvector · ChromaDB

#0360

TRADITIONAL

INTERMEDIATE

Virtual Concierge Agent

Travel & Hospitality

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over virtual concierge, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [OpenAI Agents SDK](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA Neo4j · Pinecone

#0361

TRADITIONAL

INTERMEDIATE

Demand Forecasting Copilot

Travel & Hospitality

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over demand forecasting, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0362

TRADITIONAL

INTERMEDIATE

Multilingual Support Pipeline

Travel & Hospitality

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over multilingual support, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [FAISS](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA DuckDB · Qdrant

#0363

TRADITIONAL

INTERMEDIATE

Claims Automation Knowledge Base

Insurance

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over claims automation, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA LanceDB · Milvus

#0364

TRADITIONAL

INTERMEDIATE

Risk Assessment Workflow Automation

Insurance

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over risk assessment, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Qdrant](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA OpenSearch · pgvector

#0365

TRADITIONAL

INTERMEDIATE

Policy Comparison Evaluation Harness

Insurance

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over policy comparison, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA FAISS · Neo4j

#0366

TRADITIONAL

INTERMEDIATE

Damage Estimation API Service

Insurance

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over damage estimation, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [MLflow](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0367

TRADITIONAL

INTERMEDIATE

Fraud Detection RAG Assistant

Insurance

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over fraud detection, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA Pinecone · DuckDB

#0368

TRADITIONAL

INTERMEDIATE

Underwriting Copilot Agent

Insurance

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over underwriting copilot, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Docker](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA Weaviate · LanceDB

#0369

TRADITIONAL

INTERMEDIATE

Code Review Copilot

Software Engineering

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over code review, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA Qdrant · OpenSearch

#0370

TRADITIONAL

INTERMEDIATE

Test Generation Pipeline

Software Engineering

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over test generation, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [LangFuse](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA Milvus · FAISS

#0371

TRADITIONAL

INTERMEDIATE

Incident Root Cause Knowledge Base

Software Engineering

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over incident root cause, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA pgvector · ChromaDB

#0372

TRADITIONAL

INTERMEDIATE

Documentation Workflow Automation

Software Engineering

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over documentation, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS

LangGraph

LangChain

Pydantic AI

Qdrant

LangSmith

VECTOR / INFRA

Neo4j · Pinecone

#0373

TRADITIONAL

INTERMEDIATE

Legacy Migration Evaluation Harness

Software Engineering

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over legacy migration, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS

LlamaIndex

OpenAI Agents SDK

pgvector

VECTOR / INFRA

Elasticsearch · Weaviate

#0374

TRADITIONAL

INTERMEDIATE

Developer Copilot API Service

Software Engineering

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over developer copilot, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS

DSPy

LlamaIndex

PEFT / LoRA

MLflow

Pydantic

VECTOR / INFRA

DuckDB · Qdrant

#0375

TRADITIONAL

INTERMEDIATE

Clinical Notes RAG Assistant

Healthcare

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over clinical notes, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA LanceDB · Milvus

#0376

TRADITIONAL

INTERMEDIATE

Diagnostic Imaging Agent

Healthcare

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over diagnostic imaging, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [ChromaDB](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA OpenSearch · pgvector

#0377

TRADITIONAL

INTERMEDIATE

Patient Triage Copilot

Healthcare

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over patient triage, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA FAISS · Neo4j

#0378

TRADITIONAL

INTERMEDIATE

Drug Interaction Checker Pipeline

Healthcare

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over drug interaction checker, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [pgvector](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0379

TRADITIONAL

INTERMEDIATE

EHR Records Knowledge Base

Healthcare

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over ehr records, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA Pinecone · DuckDB

#0380

TRADITIONAL

INTERMEDIATE

Remote Patient Monitoring Workflow Automation

Healthcare

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over remote patient monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [FastAPI](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA Weaviate · LanceDB

#0381

TRADITIONAL

INTERMEDIATE

Fraud Detection Evaluation Harness

Finance & Banking

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over fraud detection, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA Qdrant · OpenSearch

#0382

TRADITIONAL

INTERMEDIATE

Credit Scoring API Service

Finance & Banking

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over credit scoring, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [LangSmith](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA Milvus · FAISS

#0383

TRADITIONAL

INTERMEDIATE

Portfolio Analytics RAG Assistant

Finance & Banking

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over portfolio analytics, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA pgvector · ChromaDB

#0384

TRADITIONAL

INTERMEDIATE

Loan Underwriting Agent

Finance & Banking

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over loan underwriting, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Pydantic](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA Neo4j · Pinecone

#0385

TRADITIONAL

INTERMEDIATE

Transaction Reconciliation Copilot

Finance & Banking

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over transaction reconciliation, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0386

TRADITIONAL

INTERMEDIATE

Market Sentiment Pipeline

Finance & Banking

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over market sentiment, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [OpenAI Agents SDK](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA DuckDB · Qdrant

#0387

TRADITIONAL

INTERMEDIATE

Product Recommendations Knowledge Base

Retail & E-commerce

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over product recommendations, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA LanceDB · Milvus

#0388

TRADITIONAL

INTERMEDIATE

Demand Forecasting Workflow Automation

Retail & E-commerce

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over demand forecasting, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [FAISS](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA OpenSearch · pgvector

#0389

TRADITIONAL

INTERMEDIATE

Visual Search Evaluation Harness

Retail & E-commerce

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over visual search, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA FAISS · Neo4j

#0390

TRADITIONAL

INTERMEDIATE

Dynamic Pricing API Service (v2)

Retail & E-commerce

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over dynamic pricing, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Qdrant](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0391

TRADITIONAL

INTERMEDIATE

Cart Abandonment RAG Assistant

Retail & E-commerce

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over cart abandonment, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA Pinecone · DuckDB

#0392

TRADITIONAL

INTERMEDIATE

Inventory Optimization Agent

Retail & E-commerce

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over inventory optimization, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [MLflow](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Weaviate · LanceDB

#0393

TRADITIONAL

INTERMEDIATE

Adaptive Learning Paths Copilot

Education

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over adaptive learning paths, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LlamaIndex FastAPI Pydantic AI
VECTOR / INFRA	Qdrant · OpenSearch

#0394

TRADITIONAL

INTERMEDIATE

Essay Grading Pipeline

Education

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over essay grading, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	DSPy LlamaIndex Docker OpenAI Agents SDK pgvector
VECTOR / INFRA	Milvus · FAISS

#0395

TRADITIONAL

INTERMEDIATE

Tutoring Assistant Knowledge Base

Education

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over tutoring assistant, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LangChain LangSmith PEFT / LoRA
VECTOR / INFRA	pgvector · ChromaDB

#0396

TRADITIONAL

INTERMEDIATE

Curriculum Generator Workflow Automation

Education

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over curriculum generator, exposed through a typed API with tracing, caching and evaluation.

LLM	Claude 3.5 Sonnet
UI	React + TanStack Query
BACKEND	Python · FastAPI + SQLAlchemy 2.0 + Alembic
AI FRAMEWORKS	LangGraph LangChain LangFuse FAISS FastAPI
VECTOR / INFRA	Neo4j · Pinecone

#0397

TRADITIONAL

INTERMEDIATE

Student Engagement Analytics Evaluation Harness

Education

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over student engagement analytics, exposed through a typed API with tracing, caching and evaluation.

LLM	Gemini 1.5 Pro
UI	React + Vite + Tailwind
BACKEND	Python · FastAPI + MinIO / S3
AI FRAMEWORKS	LlamaIndex Pydantic ChromaDB
VECTOR / INFRA	Elasticsearch · Weaviate

#0398

TRADITIONAL

INTERMEDIATE

Language Learning API Service

Education

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over language learning, exposed through a typed API with tracing, caching and evaluation.

LLM	Llama 3.3 70B
UI	Streamlit + FastAPI
BACKEND	Python · FastAPI + APScheduler
AI FRAMEWORKS	DSPy LlamaIndex Pydantic AI Qdrant LangSmith
VECTOR / INFRA	DuckDB · Qdrant

#0399

TRADITIONAL

INTERMEDIATE

Predictive Maintenance RAG Assistant

Manufacturing

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over predictive maintenance, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA LanceDB · Milvus

#0400

TRADITIONAL

INTERMEDIATE

Quality Inspection Agent

Manufacturing

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over quality inspection, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [PEFT / LoRA](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA OpenSearch · pgvector

#0401

TRADITIONAL

INTERMEDIATE

Supply Chain Planning Copilot

Manufacturing

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over supply chain planning, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA FAISS · Neo4j

#0402

TRADITIONAL

INTERMEDIATE

Robotic Process Control Pipeline

Manufacturing

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over robotic process control, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [ChromaDB](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0403

TRADITIONAL

INTERMEDIATE

Defect Detection Knowledge Base

Manufacturing

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over defect detection, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA Pinecone · DuckDB

#0404

TRADITIONAL

INTERMEDIATE

Factory Digital Twin Workflow Automation

Manufacturing

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over factory digital twin, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [pgvector](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA Weaviate · LanceDB

#0405

TRADITIONAL

INTERMEDIATE

Route Optimization Evaluation Harness

Logistics & Supply Chain

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over route optimization, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA Qdrant · OpenSearch

#0406

TRADITIONAL

INTERMEDIATE

Fleet Management API Service

Logistics & Supply Chain

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over fleet management, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [FastAPI](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA Milvus · FAISS

#0407

TRADITIONAL

INTERMEDIATE

Warehouse Robotics RAG Assistant

Logistics & Supply Chain

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over warehouse robotics, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [Docker](#) [OpenAI Agents SDK](#)

VECTOR / INFRA pgvector · ChromaDB

#0408

TRADITIONAL

INTERMEDIATE

Last-Mile Delivery Agent

Logistics & Supply Chain

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over last-mile delivery, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [LangSmith](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA Neo4j · Pinecone

#0409

TRADITIONAL

INTERMEDIATE

Freight Document Processing Copilot

Logistics & Supply Chain

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over freight document processing, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0410

TRADITIONAL

INTERMEDIATE

Inventory Replenishment Pipeline

Logistics & Supply Chain

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over inventory replenishment, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Pydantic](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA DuckDB · Qdrant

#0411

TRADITIONAL

INTERMEDIATE

Threat Detection Knowledge Base

Cybersecurity

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over threat detection, exposed through a typed API with tracing, caching and evaluation.

LLM	Qwen 2.5 72B
UI	React + shadcn/ui
BACKEND	Python · FastAPI + PostgreSQL + pgvector
AI FRAMEWORKS	LangChain Pydantic AI Qdrant
VECTOR / INFRA	LanceDB · Milvus

#0412

TRADITIONAL

INTERMEDIATE

Phishing Analysis Workflow Automation

Cybersecurity

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over phishing analysis, exposed through a typed API with tracing, caching and evaluation.

LLM	Mistral Large
UI	Next.js + Tailwind
BACKEND	Python · FastAPI + Celery + Redis
AI FRAMEWORKS	LangGraph LangChain OpenAI Agents SDK pgvector LangFuse
VECTOR / INFRA	OpenSearch · pgvector

#0413

TRADITIONAL

INTERMEDIATE

Vulnerability Triage Evaluation Harness

Cybersecurity

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over vulnerability triage, exposed through a typed API with tracing, caching and evaluation.

LLM	GPT-4o
UI	Streamlit + WebSocket streaming
BACKEND	Python · FastAPI + WebSockets
AI FRAMEWORKS	LlamaIndex PEFT / LoRA MLflow
VECTOR / INFRA	FAISS · Neo4j

#0414

TRADITIONAL

INTERMEDIATE

Security Log Analysis API Service

Cybersecurity

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over security log analysis, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [FAISS](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0415

TRADITIONAL

INTERMEDIATE

Incident Response RAG Assistant

Cybersecurity

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over incident response, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA Pinecone · DuckDB

#0416

TRADITIONAL

INTERMEDIATE

Malware Classification Agent

Cybersecurity

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over malware classification, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Qdrant](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA Weaviate · LanceDB

#0417

TRADITIONAL

INTERMEDIATE

Campaign Copy Generation Copilot

Marketing & Advertising

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over campaign copy generation, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA Qdrant · OpenSearch

#0418

TRADITIONAL

INTERMEDIATE

Audience Segmentation Pipeline

Marketing & Advertising

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over audience segmentation, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [MLflow](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Milvus · FAISS

#0419

TRADITIONAL

INTERMEDIATE

Ad Creative Optimization Knowledge Base

Marketing & Advertising

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over ad creative optimization, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA pgvector · ChromaDB

#0420

TRADITIONAL

INTERMEDIATE

SEO Content Planning Workflow Automation

Marketing & Advertising

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over seo content planning, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Docker](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA Neo4j · Pinecone

#0421

TRADITIONAL

INTERMEDIATE

Social Listening Evaluation Harness · Multi-Tenant

Marketing & Advertising

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over social listening, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0422

TRADITIONAL

INTERMEDIATE

Attribution Modeling API Service · High-Availability

Marketing & Advertising

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over attribution modeling, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [LangFuse](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA DuckDB · Qdrant

#0423

TRADITIONAL

INTERMEDIATE

Ticket Classification RAG Assistant · Self-Healing

Customer Support

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over ticket classification, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA LanceDB · Milvus

#0424

TRADITIONAL

INTERMEDIATE

Virtual Support Agent Agent · Observable

Customer Support

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over virtual support agent, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Pydantic AI](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA OpenSearch · pgvector

#0425

TRADITIONAL

INTERMEDIATE

Sentiment Routing Copilot · Enterprise

Customer Support

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over sentiment routing, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA FAISS · Neo4j

#0426

TRADITIONALINTERMEDIATE

Knowledge Base Search Pipeline · Real-Time

Customer Support

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over knowledge base search, exposed through a typed API with tracing, caching and evaluation.

LLM	Claude 3.5 Sonnet
UI	React + TanStack Query
BACKEND	Python · FastAPI + SQLAlchemy 2.0 + Alembic
AI FRAMEWORKS	DSPyLlamaIndexPEFT / LoRAMLflowPydantic
VECTOR / INFRA	ChromaDB · Elasticsearch

#0427

TRADITIONALINTERMEDIATE

Escalation Prediction Knowledge Base · Multilingual

Customer Support

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over escalation prediction, exposed through a typed API with tracing, caching and evaluation.

LLM	Gemini 1.5 Pro
UI	React + Vite + Tailwind
BACKEND	Python · FastAPI + MinIO / S3
AI FRAMEWORKS	LangChainFAISSFastAPI
VECTOR / INFRA	Pinecone · DuckDB

#0428

TRADITIONALINTERMEDIATE

Voice Support Bot Workflow Automation · Edge-Optimized

Customer Support

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over voice support bot, exposed through a typed API with tracing, caching and evaluation.

LLM	Llama 3.3 70B
UI	Streamlit + FastAPI
BACKEND	Python · FastAPI + APScheduler
AI FRAMEWORKS	LangGraphLangChainChromaDBDockerOpenAI Agents SDK
VECTOR / INFRA	Weaviate · LanceDB

#0429

TRADITIONAL

INTERMEDIATE

Contract Review Evaluation Harness · Privacy-Preserving

Legal & Compliance

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over contract review, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LlamaIndex](#) [Qdrant](#) [LangSmith](#)

VECTOR / INFRA Qdrant · OpenSearch

#0430

TRADITIONAL

INTERMEDIATE

Legal Research API Service · Compliance-Ready

Legal & Compliance

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over legal research, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [pgvector](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA Milvus · FAISS

#0431

TRADITIONAL

INTERMEDIATE

Policy Compliance RAG Assistant · Low-Latency

Legal & Compliance

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over policy compliance, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LangChain](#) [MLflow](#) [Pydantic](#)

VECTOR / INFRA pgvector · ChromaDB

#0432

TRADITIONAL

INTERMEDIATE

E-Discovery Agent · Offline-First

Legal & Compliance

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over e-discovery, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS

LangGraph

LangChain

FastAPI

Pydantic AI

Qdrant

VECTOR / INFRA

Neo4j · Pinecone

#0433

TRADITIONAL

INTERMEDIATE

Clause Extraction Copilot · Multi-Tenant

Legal & Compliance

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over clause extraction, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS

LlamaIndex

Docker

OpenAI Agents SDK

VECTOR / INFRA

Elasticsearch · Weaviate

#0434

TRADITIONAL

INTERMEDIATE

Regulatory Monitoring Pipeline · High-Availability

Legal & Compliance

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over regulatory monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS

DSPy

LlamaIndex

LangSmith

PEFT / LoRA

MLflow

VECTOR / INFRA

DuckDB · Qdrant

#0435

TRADITIONAL

INTERMEDIATE

Resume Screening Knowledge Base · Self-Healing

Human Resources

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over resume screening, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [LangFuse](#) [FAISS](#)

VECTOR / INFRA LanceDB · Milvus

#0436

TRADITIONAL

INTERMEDIATE

Interview Assistant Workflow Automation · Observable

Human Resources

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over interview assistant, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [Pydantic](#) [ChromaDB](#) [Docker](#)

VECTOR / INFRA OpenSearch · pgvector

#0437

TRADITIONAL

INTERMEDIATE

Employee Onboarding Evaluation Harness · Enterprise

Human Resources

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over employee onboarding, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [Pydantic AI](#) [Qdrant](#)

VECTOR / INFRA FAISS · Neo4j

#0438

TRADITIONAL

INTERMEDIATE

Attrition Prediction API Service · Real-Time

Human Resources

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over attrition prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [OpenAI Agents SDK](#) [pgvector](#) [LangFuse](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0439

TRADITIONAL

INTERMEDIATE

Skills Gap Analysis RAG Assistant · Multilingual

Human Resources

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over skills gap analysis, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LangChain](#) [PEFT / LoRA](#) [MLflow](#)

VECTOR / INFRA Pinecone · DuckDB

#0440

TRADITIONAL

INTERMEDIATE

Performance Review Summarizer Agent · Edge-Optimized

Human Resources

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over performance review summarizer, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [FAISS](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA Weaviate · LanceDB

#0441

TRADITIONAL

INTERMEDIATE

Crop Disease Detection Copilot · Privacy-Preserving

Agriculture

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over crop disease detection, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS LlamaIndex ChromaDB Docker

VECTOR / INFRA Qdrant · OpenSearch

#0442

TRADITIONAL

INTERMEDIATE

Yield Prediction Pipeline · Compliance-Ready

Agriculture

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over yield prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS DSPy LlamaIndex Qdrant LangSmith PEFT / LoRA

VECTOR / INFRA Milvus · FAISS

#0443

TRADITIONAL

INTERMEDIATE

Precision Irrigation Knowledge Base · Low-Latency

Agriculture

Design a production-ready Knowledge Base with LangChain that chains retrieval, tool calls and LLM reasoning over precision irrigation, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS LangChain pgvector LangFuse

VECTOR / INFRA pgvector · ChromaDB

#0444

TRADITIONAL

INTERMEDIATE

Pest Monitoring Workflow Automation · Offline-First

Agriculture

Design a production-ready Workflow Automation with LangGraph that chains retrieval, tool calls and LLM reasoning over pest monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM Claude 3.5 Sonnet

UI React + TanStack Query

BACKEND Python · FastAPI + SQLAlchemy 2.0 + Alembic

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [MLflow](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA Neo4j · Pinecone

#0445

TRADITIONAL

INTERMEDIATE

Soil Health Analytics Evaluation Harness · Multi-Tenant

Agriculture

Design a production-ready Evaluation Harness with LlamaIndex that chains retrieval, tool calls and LLM reasoning over soil health analytics, exposed through a typed API with tracing, caching and evaluation.

LLM Gemini 1.5 Pro

UI React + Vite + Tailwind

BACKEND Python · FastAPI + MinIO / S3

AI FRAMEWORKS [LlamaIndex](#) [FastAPI](#) [Pydantic AI](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0446

TRADITIONAL

INTERMEDIATE

Farm Drone Imagery API Service · High-Availability

Agriculture

Design a production-ready API Service with DSPy that chains retrieval, tool calls and LLM reasoning over farm drone imagery, exposed through a typed API with tracing, caching and evaluation.

LLM Llama 3.3 70B

UI Streamlit + FastAPI

BACKEND Python · FastAPI + APScheduler

AI FRAMEWORKS [DSPy](#) [LlamaIndex](#) [Docker](#) [OpenAI Agents SDK](#) [pgvector](#)

VECTOR / INFRA DuckDB · Qdrant

#0447

TRADITIONAL

INTERMEDIATE

Grid Load Forecasting RAG Assistant · Self-Healing

Energy & Utilities

Design a production-ready RAG Assistant with LangChain that chains retrieval, tool calls and LLM reasoning over grid load forecasting, exposed through a typed API with tracing, caching and evaluation.

LLM Qwen 2.5 72B

UI React + shadcn/ui

BACKEND Python · FastAPI + PostgreSQL + pgvector

AI FRAMEWORKS [LangChain](#) [LangSmith](#) [PEFT / LoRA](#)

VECTOR / INFRA LanceDB · Milvus

#0448

TRADITIONAL

INTERMEDIATE

Renewable Output Prediction Agent · Observable

Energy & Utilities

Design a production-ready Agent with LangGraph that chains retrieval, tool calls and LLM reasoning over renewable output prediction, exposed through a typed API with tracing, caching and evaluation.

LLM Mistral Large

UI Next.js + Tailwind

BACKEND Python · FastAPI + Celery + Redis

AI FRAMEWORKS [LangGraph](#) [LangChain](#) [LangFuse](#) [FAISS](#) [FastAPI](#)

VECTOR / INFRA OpenSearch · pgvector

#0449

TRADITIONAL

INTERMEDIATE

Asset Health Monitoring Copilot · Enterprise

Energy & Utilities

Design a production-ready Copilot with LlamaIndex that chains retrieval, tool calls and LLM reasoning over asset health monitoring, exposed through a typed API with tracing, caching and evaluation.

LLM GPT-4o

UI Streamlit + WebSocket streaming

BACKEND Python · FastAPI + WebSockets

AI FRAMEWORKS [LlamaIndex](#) [Pydantic](#) [ChromaDB](#)

VECTOR / INFRA FAISS · Neo4j

Energy Trading Pipeline · Real-Time

Energy & Utilities

Design a production-ready Pipeline with DSPy that chains retrieval, tool calls and LLM reasoning over energy trading, exposed through a typed API with tracing, caching and evaluation.

LLM	Claude 3.5 Sonnet
UI	React + TanStack Query
BACKEND	Python · FastAPI + SQLAlchemy 2.0 + Alembic
AI FRAMEWORKS	DSPy LlamaIndex Pydantic AI Qdrant LangSmith
VECTOR / INFRA	ChromaDB · Elasticsearch

1000 AI Projects · Frameworks Edition · 180 entries
Master in AI Engineering · Created By Balawant M Kadam